

pharmaceuticals. IoT devices track goods during transit, and blockchain ensures that this data—such as location, condition, or tampering—is securely stored and unchangeable. This helps prevent fraud, such as counterfeit medicines or spoiled food. As blockchain markets are expected to grow significantly, this technology is proving to be a game-changer for creating secure and reliable IoT systems.

Digital twins: A digital twin is a virtual copy of a physical object or system. Imagine having a digital version of your car, home, or even an entire factory that lets you monitor and manage the real thing without touching it. This technology helps track, analyse, and optimise real-world devices or systems in real-time, making it useful in industries like manufacturing and smart cities.

Factories use digital twins of machines to predict maintenance needs. For example, if a machine's digital twin shows early signs of wear, engineers can repair it before it breaks down. This reduces unexpected downtime and lowers repair costs. Bosch, a leading global manufacturer of automotive components and industrial products, uses digital twins to monitor its production lines, identifying small issues before they escalate into larger problems. It's like having a virtual mechanic watching your car around the clock.

Digital twins are also transforming urban planning. A digital replica of a building or a neighbourhood can help city planners optimise energy use, enhance safety, and manage resources more effectively. For instance, in Bengaluru, digital twins of buildings are being used to improve energy efficiency and reduce carbon emissions. They allow city officials to solve problems proactively, like repairing a bridge when its virtual model shows signs of stress before it becomes unsafe.

Digital twins provide a cost-effective way to improve efficiency and sustainability. They allow businesses and governments to test changes or solve issues in the virtual model, avoiding expensive mistakes in the real world.

The digital twin market is expanding rapidly. According to a report by E&Y, its value was \$9 billion in 2022 and is projected to reach to \$137 billion by 2030. Industries across India are adopting this technology to boost performance, reduce costs, and improve operations.

Sustainability in IoT: Sustainability is growing in importance for both individuals and businesses. IoT is playing a key role in achieving this by helping reduce energy consumption and pollution. Smart devices like thermostats and energy meters make energy use more efficient. For example, smart thermostats automatically adjust home temperatures to prevent energy waste when no one is home. In factories, IoT sensors monitor energy use in real-time, allowing machinery to make adjustments that save energy and cut costs.

Several cities today use IoT to track air pollution with sensors placed across different areas. These sensors help authorities respond quickly when pollution levels rise. Similarly, smart electricity meters in homes provide real-time feedback, encouraging people to save energy. In some parts of India, these meters have significantly reduced electricity consumption.

A survey by PwC reveals that 81% of consumers expect businesses to be environmentally conscious in their advertising and communications, while 69% say they are doing everything possible to minimise their carbon footprint. IoT is helping companies meet this expectation by enabling greener practices. For instance, businesses are using IoT to monitor goods and minimise waste in their supply chains, reducing emissions and conserving resources. By making homes smarter, cities cleaner, and industries more efficient, IoT is turning sustainability into a practical reality—not just in India but worldwide.

Interoperability and

standardisation: One challenge with IoT is that devices from different brands

don't always work well together. For example, setting up a smart home with lights, cameras, and air conditioners from different manufacturers can be difficult if the devices can't communicate with each other. This lack of compatibility can slow the growth of IoT systems, especially as they become more complex. A McKinsey report states that 70% of companies consider device compatibility a key factor in IoT success.

Open source solutions are solving this problem by creating standardised communication protocols, which act like a common language for devices to "talk" to each other. Two popular protocols are MQTT (Message Queuing Telemetry Transport) and CoAP (Constrained Application Protocol). Think of MQTT as a WhatsApp group for devices—it helps them send short updates to each other in real-time, like a smart thermostat informing your phone about the current room temperature. CoAP, on the other hand, works like browsing a website: devices send simple requests to get specific information, such as a fitness tracker asking a server for the latest health insights. Both protocols make it easy for gadgets from different brands to share data seamlessly.

In India, smart homes often have IoT devices from companies like Xiaomi, Samsung, and Philips. With MQTT and CoAP, these devices can work together, such as a smart bulb adjusting its brightness when a security camera detects movement. These protocols provide the framework that helps the system run smoothly. This is essential for scaling IoT in areas like smart cities, agriculture, and industry.

Low-code development platforms:

In the past, developers had to write complex code to build even basic IoT solutions. Today, thanks to open source low-code platforms, building IoT systems is easier, even for those with little coding experience.

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